A direct proof of the geometric thrackle bound

Conjecture 00000001568

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Abstract

For a finite straight-edge graph in the real plane, suppose that every two distinct edges have exactly one common point. We prove that the number of edges is at most the number of vertices. The argument permits collinear vertices, intersections at endpoints, and nonincident vertices lying on edges. A complete Lean 4 proof accompanies the report.

1 Statement and conventions

The supplied English statement defines a geometric thrackle as a "straight-edge graph in the plane in which every two edges intersect exactly once," and asks for the bound $|E| \le |V|$. Both original language versions are preserved verbatim in `source/original.md`.

We use the ordinary graph convention: a finite undirected simple graph $G = (V, E)$, with distinct vertices represented by distinct points of $\mathbb{R}^2$. For $u, v \in V$, write their positions simply as u, v, and draw an edge uv as the closed segment

$$[u, v] = \{(1 - t)u + tv : 0 \le t \le 1\}.$$

An unordered edge is counted once. An isolated vertex is allowed. The hypothesis is that $[u, v] \cap [x, y]$ is a singleton for any two distinct edges uv, xy. In particular, incident edges meet only at their shared endpoint. We impose no general-position assumption and do not require a nonincident edge pair to cross in their relative interiors. Thus the proof also covers the more restrictive usual drawing convention that forbids a nonincident vertex from lying inside an edge. Parallel copies of a nondegenerate straight edge would have a whole segment in common and therefore cannot satisfy the hypothesis.

Theorem 1. *Under these hypotheses, $|E| \le |V|$.*

The reference to the numerical coefficient 1.393 in the supplied statement is background to the requested improvement. Our result is the stated stronger bound $|E| \le |V|$; no historical or priority assertion about the bounds is needed in the proof.

2 A geometric sign lemma

For vectors $r = (r_1, r_2)$ and $s = (s_1, s_2)$ define $\det(r, s) = r_1 s_2 - r_2 s_1$.

Lemma 2. *Let $v, a, b, x \in \mathbb{R}^2$, with $a \ne v$ and $b \ne v$. Suppose that*

$$[v, a] \cap [v, b] = \{v\}, \qquad [v, a] \cap [a, x] = \{a\}, \qquad [v, b] \cap [a, x] \ne \varnothing.$$

Then

$$\det(a - v, b - v)\, \det(a - v, x - v) > 0.$$

Proof. Choose $p \in [v,b] \cap [a,x]$. If $p = v$, then v belongs to $[v,a] \cap [a,x] = \{a\}$, contrary to $a \neq v$. If $p = a$, then a belongs to $[v,a] \cap [v,b] = \{v\}$, again a contradiction.

We claim $D = \det(a - v, p - v) \neq 0$. Otherwise v, a, p are collinear. Among three collinear points, one is between the other two, including the endpoints in the meaning of "between." All cases give a contradiction:

- If $a \in [v,p]$, then $[v,p] \subseteq [v,b]$, so $a \in [v,a] \cap [v,b] = \{v\}$.
- If $p \in [v,a]$, then $p \in [v,a] \cap [v,b] = \{v\}$.
- If $v \in [a,p]$, then $[a,p] \subseteq [a,x]$, so $v \in [v,a] \cap [a,x] = \{a\}$.

These contradict respectively $a \neq v$, $p \neq v$, and $a \neq v$. The subset assertions follow from the convexity of a segment.

Write $p = (1-t)v + tb = (1-s)a + sx$, where $s, t \in [0,1]$. As $p \neq v, a$, we have $t > 0$ and $s > 0$. Bilinearity and alternation of the determinant give

$$D = t \det(a - v, b - v) = s \det(a - v, x - v)\,.$$

Hence

$$0 < D^2 = st \det(a - v, b - v) \det(a - v, x - v)\,.$$

Since $st > 0$, the claimed strict inequality follows. This argument deals explicitly with possible collinearity rather than assuming it away. □

3 At most two neighbors of degree other than one

Write $d(v)$ for the degree of a vertex. A leaf is a vertex of degree one.

Lemma 3. *Every vertex has at most two neighbors that are not leaves.*

Proof. Let a be a neighbor of v that is not a leaf. Since it already has the neighbor v, its degree is at least two. Choose a neighbor $x \neq v$ of a. For any other neighbor $b \neq a$ of v, the edges va, vb, ax are pairwise distinct: equality of vb and ax would force either $v = a$ or $v = x$, both excluded. Their singleton intersections satisfy Lemma 2. Consequently,

$$\det(a - v, b - v)\, \det(a - v, x - v) > 0 \quad \text{for every such } b.$$

In particular all the determinants $\det(a - v, b - v)$, as b ranges over the other neighbors of v, have the same strict sign. The same conclusion holds for each nonleaf neighbor in place of a.

Suppose three distinct nonleaf neighbors a, b, c exist, and put

$$A = \det(a - v, b - v)\,, \qquad B = \det(a - v, c - v)\,, \qquad C = \det(b - v, c - v)\,.$$

The conclusion at a gives $AB > 0$. At b, determinant antisymmetry gives $(-A)C > 0$. At c it gives $(-B)(-C) = BC > 0$. The first and third inequalities say that A, B, C have the same strict sign, whereas the second says that A, C have opposite signs. This contradiction proves the lemma. □

4 Counting the edges

Proof of Theorem 1. Let L be the set of leaves, let $\ell(v)$ be the number of leaf neighbors of v, and let $\mathbf{1}_L(v)$ be one when $v \in L$ and zero otherwise. Lemma 3 gives $d(v) \leq 2 + \ell(v)$ for every

vertex. If v itself is a leaf then $d(v) + \mathbf{1}_L(v) = 2 \leq 2 + \ell(v)$; otherwise the indicator is zero. Thus, in all cases,

$$d(v) + \mathbf{1}_L(v) \leq 2 + \ell(v).$$

Summing over vertices and reversing the order in the finite adjacency sum gives

$$\sum_{v \in V} \ell(v) = \sum_{w \in L} \#\{v \in V : vw \in E\} = \sum_{w \in L} d(w) = |L|.$$

The degree-sum formula therefore yields

$$2|E| + |L| \leq 2|V| + |L|.$$

Canceling $|L|$ and dividing by two proves $|E| \leq |V|$. This includes the empty graph, isolated vertices, disconnected graphs, and edges whose two endpoints are both leaves. □

5 Formal statement and verification

The project uses Lean `4.19.0` and stock Mathlib at revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

The committed Lake manifest pins the complete dependency set.

The structure `GeometricThrackle.Drawing G` contains an injective position map into $\mathbb{R} \times \mathbb{R}$, and an existence-and-uniqueness assertion for the intersection of each pair of distinct unordered edges. The custom segment definition is proved equal to Mathlib's real closed segment in `seg_eq_segment`. The final declaration is

`GeometricThrackle.geometric_thrackle_bound`

and its conclusion is

`G.edgeFinset.card ≤ Fintype.card V.`

The finite vertex type and simple graph are arbitrary. The decidable adjacency instance appearing in the Lean statement provides finite enumeration; it is available classically for every such graph and adds no mathematical restriction.

The source files correspond directly to the preceding proof:

- `Geometry.lean`: the segment definition, collinearity step, and Lemma 2 (`det_strict_side`).
- `Angular.lean`: the sign contradiction and the cardinal bound on nonleaf neighbors.
- `Counting.lean`: the finite-sum cancellation proving the edge bound from the local neighbor bound.
- `Main.lean`: the exact drawing hypotheses, their application to the geometric lemma, and the final theorem.

The verification script rebuilds all four local modules, checks the pinned manifest and toolchain, scans the authored Lean sources for incomplete or unsafe proof mechanisms, and checks the printed axioms of the final theorem. The printed dependencies are precisely `propext`, `Classical.choice`, and `Quot.sound`, the standard foundational axioms used by Mathlib. There are no added axioms, admitted goals, or computation-based trust extensions. No auxiliary numerical computation is required by the proof.

The proof was derived from the supplied statement and repository rules, with access to stock Mathlib and the pinned toolchain. No prior candidate-specific solution or online mathematical proof was consulted. The local provenance notes record the independent geometric assessment and the division of formalization work. This report makes no claim of historical novelty.